Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Arlinda Holsteins

Physical address of dairy:

6325 West main ST	Turlock	Stanislaus	95380
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address): Faithhome road

Date facility was originally placed in operation: 02/23/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X023-X040-X001-XXXX

B. OPERATORS

Vieira, Carl

Operator name: Vieira, Carl

Telephone no.: (209) 667-4644 (Landline) (Cellular)

P.O. Box 3308	Turlock	CA	95381
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Vieira, Carl

Legal owner name: Vieira, Carl

Telephone no.: (209) 667-4644 (Landline) (Cellular)

P.O. Box 3308	Turlock	CA	95381
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	60	20	150	160	110	225
Number under roof	280	0	0	0	0	0
Maximum number	340	20	160	160	110	225
Average number	320	20	160	160	110	225
Avg live weight (lbs)	1,400	1,450	600	600		

Predominant milk cow breed: Holstein

Average milk production: 83 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 13,078.16 *tons per reporting period*

Total nitrogen from manure: 164,841.49 *lbs per reporting period*

After ammonia losses (30% loss applied): 115,389.04 *lbs per reporting period*

Total phosphorus from manure: 26,174.15 *lbs per reporting period*

Total potassium from manure: 64,016.07 *lbs per reporting period*

Total salt from manure: 155,271.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,055,275 *gallons*

Total nitrogen generated: 22,796.24 *lbs*

Total phosphorus generated: 2,227.76 *lbs*

Total potassium generated: 25,698.53 *lbs*

Total salt generated: 98,106.21 *lbs*

	4,055,275	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,055,275	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID Lateral 4	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/28/2023	Solid commercial fertilizer *UN32*	1.35 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	863.14	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	863.14	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	Corral solids	730.00 *ton*	Dry-weight	32.8		15,400.00	1,500.00	11,800.00		56.00
10/02/2023	Corral solids	690.00 *ton*	Dry-weight	10.7		24,100.00	5,470.00	37,800.00		38.10

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	44,808.64	8,212.58	58,159.67	1,018,948.74
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	44,808.64	8,212.58	58,159.67	1,018,948.74

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Arlinda	39	39	2	process wastewater	X023-X040-X001-XXXX
Hatch	27	27	2	manure	X058-X010-X036-XXXX
Totals for areas that were used for application	66	66	4		
Totals for areas that were not used for application					
Land application area totals	66	66	4		

B. CROPS AND HARVESTS

Arlinda

Field name: Arlinda

11/18/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 39 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/17/2023	531.00 *ton*	As-is		70.0	6,900.00	1,000.00	17,100.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	216.00	30.00	196.80	0.00
Total actual harvest content	13.62	187.89	27.23	465.65	841.43

04/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 04/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/31/2023	1,125.00 *ton*	As-is		70.3	4,010.00	720.00	6,380.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	28.85	231.35	41.54	368.08	1,182.29

Hatch

Hatch

Field name: Hatch

11/18/2002: Ryegrass Silage

Crop: Ryegrass Silage Acres planted: 27 Plant date: 11/18/2002

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/17/2023	241.00 *ton*	As-is		70.0	13,600.00	1,500.00	17,700.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	216.00	30.00	196.80	0.00
Total actual harvest content	8.93	242.79	26.78	315.98	669.44

05/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/31/2023	698.00 *ton*	As-is		72.9	3,710.00	550.00	4,720.00		6.61

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.85	191.82	28.44	244.04	926.17

NUTRIENT BUDGET

A. LAND APPLICATIONS

Arlinda - 11/18/2022: Rye Grass Silage

Field name: Arlinda

Crop: Rye Grass Silage — Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.43	1.16	72.41	210.82	586,454.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	0.00	5,265,000.00 *gal*
Application event totals		46.43	1.16	72.41	210.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	108.98	18.37	116.11	430.45	640,670.00 *gal*
Application event totals		108.98	18.37	116.11	430.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.33	17.58	111.15	412.06	613,293.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	27.04	4,860,000.00 *gal*
Application event totals		104.33	17.58	111.15	439.10	

Arlinda - 04/30/2023: Corn, silage

Field name: Arlinda

Crop: Corn, silage — Plant date: 04/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Arlinda - 04/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	147.51	4.25	113.32	498.55	534,393.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	31.54	5,670,000.00 *gal*
Application event totals		147.51	4.25	113.32	530.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	27.04	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	28.16	5,062,500.00 *gal*
Application event totals		0.00	0.00	0.00	28.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	27.04	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.09	5.25	81.98	321.22	560,155.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	24.78	4,455,000.00 *gal*
Application event totals		59.09	5.25	81.98	346.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	24.78	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.78	

Arlinda - 04/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.09	5.25	81.98	321.22	560,155.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	22.53	4,050,000.00 *gal*
Application event totals		59.09	5.25	81.98	343.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.09	5.25	81.98	321.22	560,155.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	21.40	3,847,500.00 *gal*
Application event totals		59.09	5.25	81.98	342.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	20.28	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.28	

Hatch - 11/18/2002: Ryegrass Silage

Field name: Hatch

Crop: Ryegrass Silage

Plant date: 11/18/2002

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	279.10	83.35	369.29	2,487.25	160.00 *ton*
Application event totals		279.10	83.35	369.29	2,487.25	

Hatch - 11/18/2002: Ryegrass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	0.00	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	32.55	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.55	

Hatch - 05/01/2023: Corn, silage

Field name: Hatch

Crop: Corn, silage

Plant date: 05/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	222.31	21.65	170.34	8,083.91	290.00 *ton*
Application event totals		222.31	21.65	170.34	8,083.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	39.05	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Hatch - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	32.55	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	35.80	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	35.80	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	32.55	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	29.29	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	29.29	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Hatch - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	29.29	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

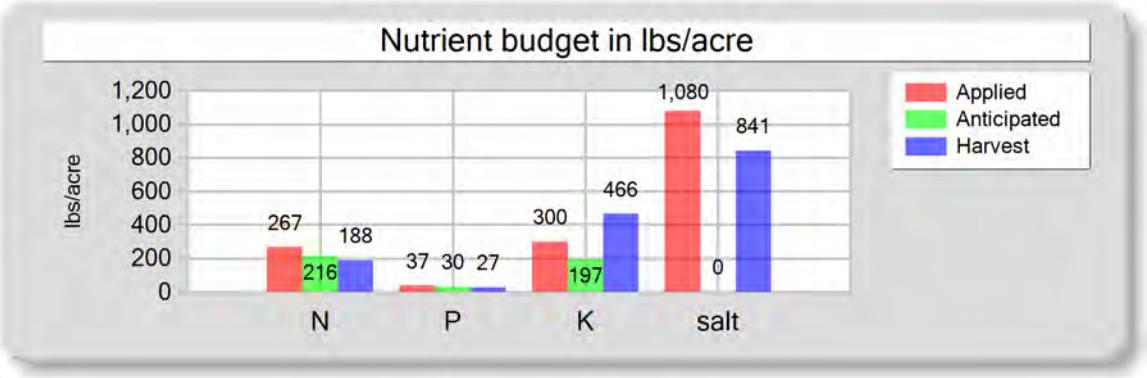
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	29.29	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

B. NUTRIENT BUDGET

Arlinda - 11/18/2022: Rye Grass Silage

Field name: Arlinda | Crop: Rye Grass Silage | Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	259.74	37.12	299.67	1,053.33
Fresh water	0.00	0.00	0.00	27.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.74	37.12	299.67	1,080.37
Anticipated crop nutrient removal	216.00	30.00	196.80	0.00
Actual crop nutrient removal	187.89	27.23	465.65	841.43
Nutrient balance	78.85	9.89	-165.98	238.94
Applied to removed ratio	1.42	1.36	0.64	1.28

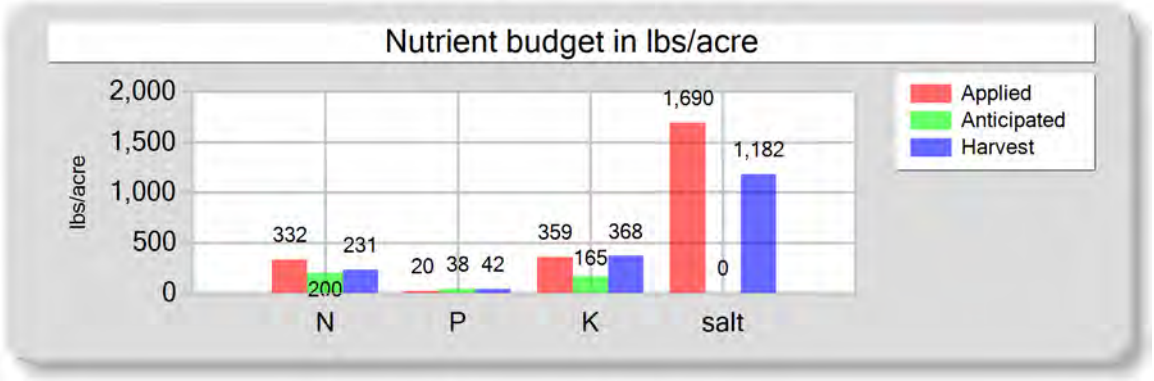
Fresh water applied	
10,125,000.00	*gallons*
372.87	*acre-inches*
9.56	*inches/acre*

Process wastewater applied	
1,840,417.00	*gallons*
67.78	*acre-inches*
1.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

Arlinda - 04/30/2023: Corn, silage

Field name: Arlinda | Crop: Corn, silage | Plant date: 04/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	324.78	20.00	359.27	1,462.22
Fresh water	0.00	0.00	0.00	227.57
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	331.78	20.00	359.27	1,689.78
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	231.35	41.54	368.08	1,182.29
Nutrient balance	100.43	-21.54	-8.81	507.49
Applied to removed ratio	1.43	0.48	0.98	1.43

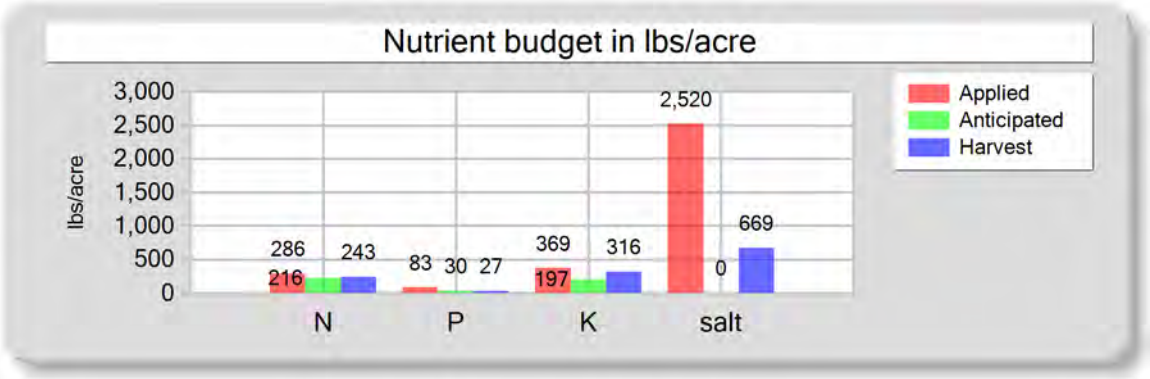
Fresh water applied	
40,905,000.00	*gallons*
1,506.39	*acre-inches*
38.63	*inches/acre*

Process wastewater applied	
2,214,858.00	*gallons*
81.57	*acre-inches*
2.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hatch - 11/18/2002: Ryegrass Silage

Field name: Hatch Crop: Ryegrass Silage Plant date: 11/18/2002



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	279.10	83.35	369.29	2,487.25
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	32.55
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.10	83.35	369.29	2,519.80
Anticipated crop nutrient removal	216.00	30.00	196.80	0.00
Actual crop nutrient removal	242.79	26.78	315.98	669.44
Nutrient balance	43.32	56.57	53.31	1,850.35
Applied to removed ratio	1.18	3.11	1.17	3.76

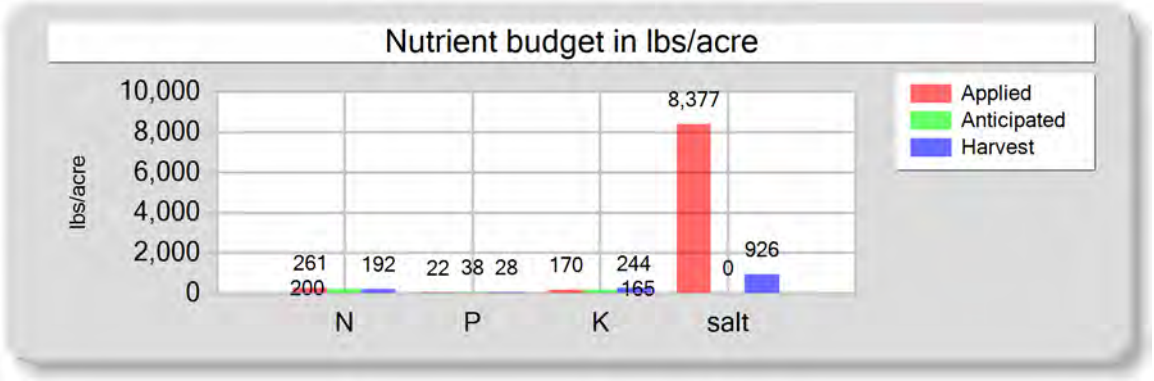
Fresh water applied	
8,505,000.00	*gallons*
313.21	*acre-inches*
11.60	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hatch - 05/01/2023: Corn, silage

Field name: Hatch | Crop: Corn, silage | Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	222.31	21.65	170.34	8,083.91
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	292.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.31	21.65	170.34	8,376.82
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	191.82	28.44	244.04	926.17
Nutrient balance	69.49	-6.78	-73.70	7,450.65
Applied to removed ratio	1.36	0.76	0.70	9.04

Fresh water applied	
36,450,000.00	*gallons*
1,342.33	*acre-inches*
49.72	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/13/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 19.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,400.00	8,780.00	38,900.00							26.20
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 32.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	1,500.00	11,800.00							56.00
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,100.00	5,470.00	37,800.00							38.10
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 10/27/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	370.00	230.00	0.00	0.00	9.28	577.00								4,930.00	1,680
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	795.00	420.00	0.00	0.00	134.00	847.00								5,770.00	3,140
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,290.00	510.00	0.00	0.00	37.20	991.00	131.00	87.70	149.00	4,480.00	0.00	259.00	158.00	6,760.00	4,360
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	493.00	328.00	0.00	0.00	43.80	684.00								5,130.00	2,680
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	512.00	305.00	0.00	0.00	23.20	915.00								6,200.00	5,020
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

TID Lateral 4

TID Lateral 4

Sample description: TID Lateral 4

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

TID Lateral 4

Sample description: TID Lateral 4

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								31.10	26
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Arlinda - 11/18/2022: Rye Grass Silage

Ryegrass Silage

Sample and source description: Ryegrass Silage

Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 45.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	1,840.00	30,900.00		10.30
DL	100.00	100.00	100.00		0.01

Arlinda - 04/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,010.00	720.00	6,380.00		6.90
DL	100.00	100.00	100.00		0.01

Hatch - 11/18/2002: Ryegrass Silage

Ryegrass

Sample and source description: Ryegrass

Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	1,950.00	23,000.00		12.50
DL	100.00	100.00	100.00		0.01

Hatch - 05/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,710.00	550.00	4,720.00		6.61
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

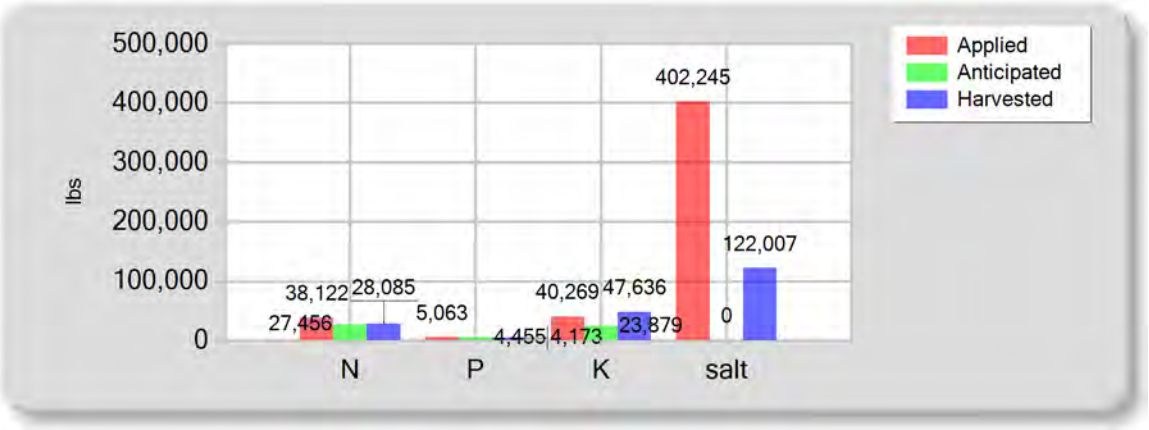
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

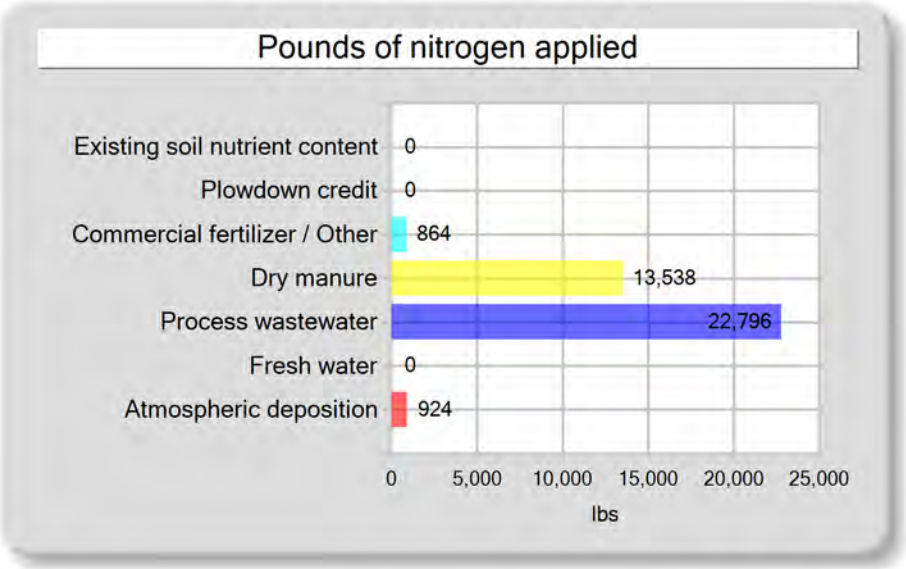
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

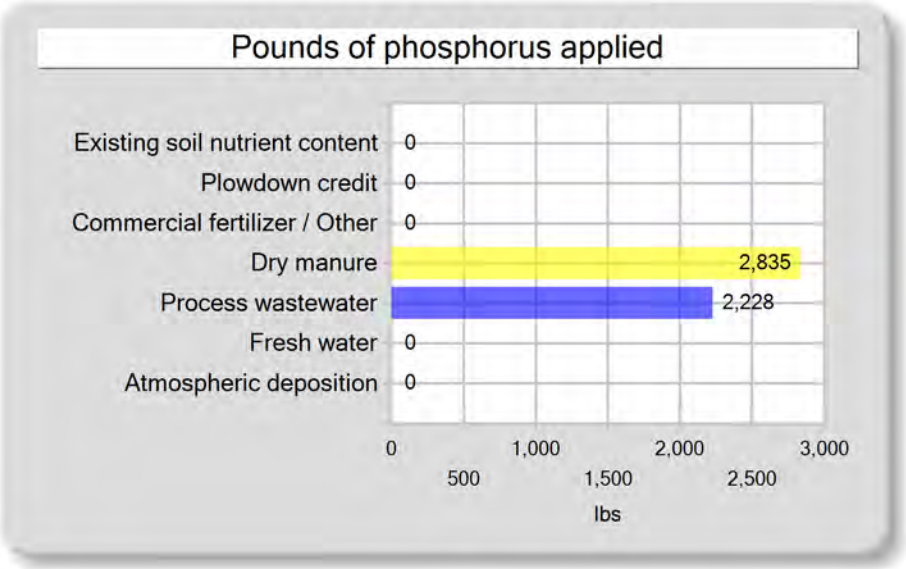
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	864.00	0.00	0.00	0.00
Dry manure	13,538.11	2,835.13	14,570.02	285,421.44
Process wastewater	22,796.24	2,227.76	25,698.53	98,106.21
Fresh water	0.00	0.00	0.00	18,716.92
Atmospheric deposition	924.00	0.00	0.00	0.00
Total nutrients applied	38,122.36	5,062.89	40,268.54	402,244.57
Anticipated crop nutrient removal	27,456.00	4,455.00	23,878.80	0.00
Actual crop nutrient removal	28,084.66	4,172.80	47,635.72	122,006.74
Nutrient balance	10,037.70	890.09	-7,367.18	280,237.83
Applied to removed ratio	1.36	1.21	0.85	3.30

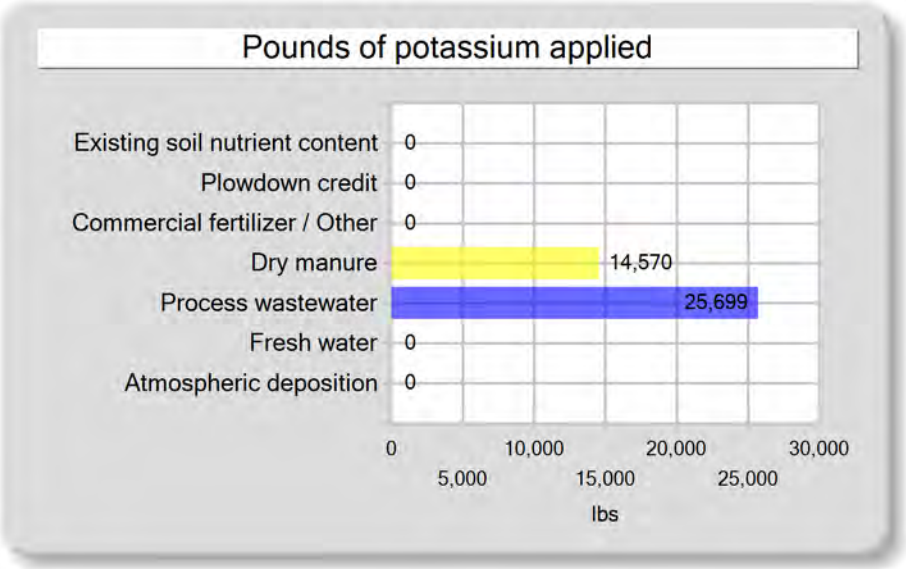
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

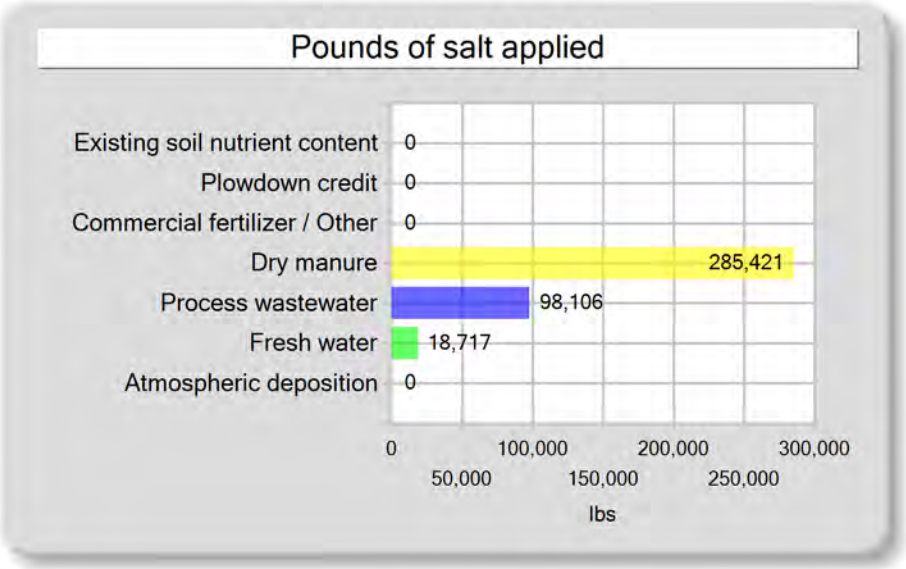


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Carl J Vieira	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Carl Vieira	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/20/24	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: CARL VIEIRA

Name of Dairy Facility: ARLINDA HOLSTEINS

Facility Address: 6325 W. MAIN ST (Number and Street) JURLOCK (City) 95380 (Zip Code)

Contact Person Name and Phone Number: CARL VIEIRA (Name) 209-667-4644 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: ALP CUSTOM SPREADING

Address of Hauling Company /Person: 2324 VILLA MANUCHA (Number and Street) NEWMAN (City) 95360 (Zip Code)

Contact Person: JEFF LIMA (Name) 209-862-3393 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

JEFF LIMA (Name) 2324 VILLA MANUCHA (Number and Street) NEWMAN (City) 95360 (Zip Code) 209-862-3393 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

2324 VILLA MANUCHA (Number and Street) NEWMAN (City) 95360 (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 4/18/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 730 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 32.6 % MOISTURE

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

73 LOADS @ 10 TONS PER LOAD

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Carl J Vierra Date: 6/20/24

Hauler's Signature: _____ Date: 4/9/24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: CARL VIEIRA

Name of Dairy Facility: ARLINDA HOLSTEINS

Facility Address: 6325 W. MAIN ST (Number and Street) JURLOCK (City) 95380 (Zip Code)

Contact Person Name and Phone Number: CARL VIEIRA (Name) 209-667-4644 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: ALP CUSTOM SPREADING

Address of Hauling Company /Person: 2324 VILLA MANUCHA (Number and Street) NEWMAN (City) 95360 (Zip Code)

Contact Person: JEFF LIMA (Name) 209-862-3393 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

JEFF LIMA (Name) 2324 VILLA MANUCHA (Number and Street) NEWMAN (City) 95360 (Zip Code) 209-862-3393 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

2324 VILLA MANUCHA (Number and Street) NEWMAN (City) 95360 (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 10/2/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 690 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 10.7% MOISTURE

Manure Density (if amount reported in cubic yards): _____

Attachment D
Waste Discharge Requirements General Order No. R5-2007-0035
Existing Milk Cow Dairies

Method used to determine amount of manure: _____

69 LOADS @ 10 TONS PER LOAD.

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Carl Vieira Date: 6/28/24

Hauler's Signature: Jeff Date: 4/9/24

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
CARL VIEIRA DAIRY
P.O. Box 3308
Turlock, CA 95381

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Arlinda Holsteins
RECEIVED DATE: 12/12/23 3:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Milk Barn	T3346035A	Electrical Conductivity (EC)	EPA 120.1 and	0.267	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	171	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	100	98.2 (T3346035A)	98.8 (T3346035A)	0.608
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)	1.40

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Arlinda Holsteins
RECEIVED DATE: 12/12/23 3:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
West Main	T3346035B	Electrical Conductivity (EC)	EPA 120.1 and	0.267	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.187	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	171	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	100	98.2 (T3346035A)	98.8 (T3346035A)	0.608
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)	1.40

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: CARL VIEIRA
Mailing address:
(not facility address) TURLOCK CA
Phone#:

Authorized Copy Released To: F+R AG
Sampled By: BILL
Project ID: ARLINDA HOLSTEINS

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12/12/23	11:00	MILK BARN	DOMESTIC	13.0°c		X				
2	"	11:15	WEST MAIN	"	11.0°c		X				
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged in By:	LABORATORY NUMBER
1st		F+R				
2nd		DENELE ANALYTICAL, INC.	12-12-23 3:00			T3346035

Required to be submitted to any agency? (YES / NO)